



DA237X – Degree Project in Computer Science and
Engineering, specialising in Cybersecurity

Individual Plan

Benchmarking LLM-driven Model evasion attacks
against NIDS

Eduardo Santos Rodríguez

KTH Royal Institute of Technology School of Electrical Engineering and
Computer Science

April 30, 2026

1 PROJECT INFORMATION

Preliminary title, that indicates what the degree project will be about. The name and e-mail address of the student The name of the examiner at KTH The name of the supervisor at KTH The name and e-mail address of the supervisor, if the thesis is performed outside KTH Current date Keywords

- **Preliminary title:** Benchmarking LLM-driven Model evasion attacks against NIDS
- **Name and email of the student:** Eduardo Santos Rodríguez. edsr@kth.se
- **KTH Examiner:** Philipp Haller. phaller@kth.se
- **Supervisor outside KTH:** Han Wang. han.wang@ri.se
- **Date:** April 30, 2026
- **Keywords:** Cybersecurity, Large Language Model, Network Intrusion Detection System

2 BACKGROUND & OBJECTIVE

Although Network Intrusion Detection Systems, or NIDS, have already been part of the defensive cybersecurity environment for many years, they continue to play an important role in real-world applications of digital infrastructure. Traditional versions based on signature analysis have a number of weaknesses that have been extensively studied [2]. These shortcomings in traditional versions, combined with the current technological boom in the field of artificial intelligence (AI), has led to the emergence of new versions of NIDS that specifically incorporate various techniques related to the field of AI [7].

At the same time, the offensive side of cybersecurity has also been disrupted by this new tool. New AI techniques are constantly being deployed in an attempt to bypass the defenses of IDS-type systems [6]. This applies to both traditional systems [4] and those that, in turn, rely on some form of AI [8].

The Research Institute of Sweden (RISE) is tasked with investigating these kinds of new developments to determine the extent to which they can be applied in practice.

The aim of this master's thesis is precisely to investigate whether a widely used AI tool, such as large language models (LLMs) can help to partially or totally bypass the defenses of NIDS systems in real-world traffic scenarios (which do not necessarily involve live traffic).

The knowledge gained through KTH's Master's program in cybersecurity provides an excellent starting point for understanding NIDS systems and testing them.

3 RESEARCH QUESTION & METHOD

3.1 Question & Hypothesis

The **hypothesis** that this master's thesis aims to validate is that by using LLMs, malicious attack traffic can be modified in such a way that the evasion/detection rate of Network Intrusion Detection Systems changes significantly.

With this in mind, the '**Research Question**' follows: To what extent can Large Language Models be used to modify malicious traffic in order to reduce the detection rate of attacks by Network Intrusion Detection Systems, compared to the number of attacks detected by the same type of systems when pre-modified traffic is used to carry out the same of attacks?

3.2 Objectives

To answer the Research Question, the thesis will create and apply an indicator, the **Signature Evasion Rate (SER)** that compares the grade of evasion achieved against the NID-system in the following situations: (1) when attack packets have not been modified, and (2) when the same traffic has been modified by different LLMs with the aim of evading detection of the attack.

Secondary metrics, such as *Accuracy*, *Sensitivity (also known as Recall)*, or *F1*, might also be used to evaluate the performance of the NIDS (Snort 3). These metrics are extensively used in scientific literature, as observed in [3].

3.3 Method

The tests will consist of comparing the results obtained after the analysis of non-modified and modified-by-the-LLMs .pcap files by the chosen Network Intrusion Detection System: **Snort 3** [7]. This tool is designed to scan these files and generate alert logs if any anomalies are detected within them. Snort 3 uses two detection mechanisms that work together to detect potential anomalies:

- **Signature-based** detection: Its functionality is based on the rules stored in a local rules set. If a packet "breaks" any of these static rules, an alarm will be triggered. This rule set can be modified by adding self-crafted rules and/or including available third-party public rules.
- **Anomaly-based** detection: This mechanism relies on the so-called *inspectors*. Snort 3 has a dedicated inspector for each communication protocol (such as HTTP, SSH or TLS). These ensure that traffic using that protocol complies with the relevant standards (RFCs). If a packet attempts to exploit a structural weakness in the protocol (for example, a malformed length field), the inspector flags it as an anomaly long before it is compared against a signature database. Unlike simple signatures, inspectors maintain the connection state. This enables them to detect anomalous behaviour in the communication sequence, which is fundamental to anomaly-based IDS.

To ensure the reproducibility and availability of the thesis, the .pcap files and the traffic/packets contained in them, which are to be used for both testing and LLM-fine-tuning purposes, are to be obtained from public traffic datasets. The currently selected dataset to be used for the **testing** phase is presented in [1]. Besides the already mentioned one, another dataset introduced by [5] could be used for the **training phase**. These datasets contain examples of both benign and malicious traffic. Any changes to the dataset selection will be justified in future documents.

The same logic applies to the selection of the rule set, which is a public set provided by Cisco and located on their [website](#). Cisco is the owner of the Snort 3 tool. It may be that the number of rules in the set is reduced so that the final version contains only rules designed to counter the attacks ultimately selected for testing. The reason for doing this is to make the tests more realistic in terms of digital infrastructure and to improve the performance of the Snort tool; as [4] demonstrates, this occurs when the rule set is fine-tuned.

The choice of attacks to be considered is currently under discussion. The idea is to support this choice with scientific documentation on the significance and frequency of the various attacks today.

Finally, the metric mentioned in 3.2 will show how effective the LLM-driven traffic modifications are. For this purpose, both the number and the content of the alerts contained in the log generated by the NIDS must be taken into account. A matrix will be used to evaluate each entry generated in the alert log that is produced following the analysis of the modified traffic, thereby determining which of these entries will be taken into account when calculating the metric. The possible outputs of the matrix are: **Successful evasion**: When an alarm present in the non-modified version of the testing-traffic is not present in the modified version of the same traffic created by the LLM; **Failed evasion**: When the SAME alarm is still present; **Partial evasion**: When a packet/packet flow generates an alarm for the unmodified traffic but generates a different alarm in the modified traffic log; **Noise**: When the alarm present in the modified traffic log bears no relation whatsoever to the alarms in the unmodified traffic log.

With regard to comparing content, the challenge is to establish a one-to-one correspondence between the alerts generated by unmodified traffic and those generated by modified traffic. In order to achieve this traceability, the idea is to restrict the modification of certain fields in the packets that make up the traffic, as done in [6], where any potential modification would be of little practical use for the main objective of the thesis. Furthermore, these must be fields that can be linked to the alerts generated by the NIDS to achieve alert-to-packet/packet-flow traceability.

3.4 Tasks

1. Pre-study: Search and analysis of relevant literature.
2. Creation of digital infrastructure to carry out the tests part I: Creation of a machine where to run the software tools, the installation and configuration of Snort 3, and the installation of Wireshark for manual packet analysis.

3. Selection of traffic datasets and LLMs to use in the thesis.
4. Fine-tuning of Snort 3.
5. Generation of alert logs with the non-modified traffic of the chosen attacks.
6. Creation of digital infrastructure to carry out the tests part II: Setting up of the selected LLMs to modified the traffic.
7. Creation of python scripts to parse the traffic from pcap format to a format like .csv or Hex that the LLM can understand and modify, as well as parsing it back to .pcap for the Snort 3 analysis.
8. Modification of the traffic files via LLM.
9. Generation of alert logs with the modified traffic of the chosen attacks.
10. Evaluation of results and metric calculation.
11. **Optional** If the LLMs used do not improve the **SER** on a significant level, these models would be then fine-tuned in order to improve their performance. This is known as **training phase**. Once this last one is completed, the testing phase (steps 5 to 10) would be re-started and the results would be again analysed.

3.5 Limitations

In this thesis, the scope will be limited to a select number of LLMs and the types of attacks contained in the selected datasets for the testing phase (See [1]). Although the evasion/detection rates will vary depending on the combination type of attack and LLMs, the aim of this thesis is not to compare all these cases, but rather to investigate whether LLMs can be used as tools for the purpose described above.

3.6 Risks

One potential outcome is that the initial modification of the LLMs fails to improve the evasion rate. In that case, the option of fine-tuning these models prior to modifying the traffic files would be considered. The modification and subsequent evaluation would then be repeated, but this time using LLMs specifically trained to bypass our NIDS. However, even after this training phase, the LLMs could be unable to increase the evasion rate.

4 EVALUATION & NEW VALUE

The fulfillment of the research objectives will be measured by the SER value. This quantitative metric compares the detection logs of Snort 3 when processing original malicious traffic from the selected datasets versus traffic modified by LLMs. Success is achieved if the LLMs generate adversarial patterns that significantly lower detection rates without breaking the underlying attack functionality.

The scientific relevance lies in establishing a systematic benchmark for LLM-driven Model Evasion Attacks (MEA) against traditional NIDS. While legacy systems rely on static signatures, this work explores how generative AI can automate polymorphic transformations to bypass these defenses. It provides a baseline for understanding how LLMs compromise established network security layers.

The new value of this research stems from its practical focus on real-world traffic scenarios. The results will be of particular interest to organizations relying on traditional NIDS (like Snort or Suricata) that need to understand the urgency of transitioning toward AI-driven or behavior-based defense mechanisms to counter the next generation of AI-automated threats.

5 PRE-STUDY

The literature related to this thesis primarily focuses on scientific papers related to the 2 subsystems the digital infrastructure of the thesis is built upon: The NIDS (defensive side) and AI-powered system used to modify the already malicious traffic (attacking side). The details of these scientific sources can be found on the [7](#) section.

6 CONDITIONS & SCHEDULE

6.1 Resources

The following items and/or software tools are needed to carry out the tests that might answer the research question:

- Snort 3. Latest version.
- Wireshark. Latest version.
- Python. Versions 3.12 - 3.14 are recommended.
- An IDE to develop the Python scripts. The latest VSCode version will be used in this thesis.

- A Virtual or Physical machine with an OS capable of running the already mentioned software tools. For this thesis, I will use a virtual machine implemented over Oracle VirtualBox. As the OS, I will be using Ubuntu 22.04.4, as it is the latest LTS version of Ubuntu available during the work on this thesis.
- A machine to run the LLMs in charge of the modifications.
- Infrastructure for training/fine-tuning LLMs.

6.2 RISE Supervisor

Han Wang will support my project by:

- Organising and conducting weekly meetings to ensure the thesis work is progressing smoothly.
- Holding ad hoc meetings or discussions to address any potential issues that arise during the course of the thesis.
- Bringing her extensive knowledge and expertise in AI, which will be essential for the section on working with LLMs.
- Acting as a liaison with RISE. In particular, with the cybersecurity group within that institution.

6.3 Schedule

Master Thesis: Benchmarking LLM-driven Model evasion attacks against NIDS								
Tasks	Months							
	Jan	Feb	Mar	Apr	May	Jun	July	August
Literature Research								
Build State-of-the-arts and Testing Design								
Implementation of Digital Infrastructure								
Testing Execution and Evaluation								
Thesis Report writing								

Figure 1: Preliminary Project Timeline

7 USE OF GENERATIVE AI

I have used/ I am planning to use AI-powered tools for the following purposes:

- Adapting and optimizing the LATEX code to redact the present document
- Improving and/or correcting the use of the English language in the thesis document or in any other document within the thesis

References

- [1] Iman Sharafaldin, Arash Habibi Lashkari, and Ali A. Ghorbani. “Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization”. In: *Proceedings of the 4th International Conference on Information Systems Security and Privacy, ICISSP 2018, Funchal, Madeira - Portugal, January 22-24, 2018*. Ed. by Paolo Mori, Steven Furnell, and Olivier Camp. SciTePress, 2018, pp. 108–116. DOI: [10.5220/0006639801080116](https://doi.org/10.5220/0006639801080116). URL: <https://doi.org/10.5220/0006639801080116>.
- [2] Hakan Kilic, Neset Sertac Katal, and Ali Aydin Selcuk. “Evasion Techniques Efficiency Over The IPS/IDS Technology”. eng. In: *2019 4th International Conference on Computer Science and Engineering (UBMK)*. IEEE, 2019, pp. 542–547. ISBN: 9781728139647.
- [3] Hanan Hindy et al. “A Taxonomy of Network Threats and the Effect of Current Datasets on Intrusion Detection Systems”. In: *IEEE Access* 8 (2020), pp. 104650–104675. DOI: [10.1109/ACCESS.2020.3000179](https://doi.org/10.1109/ACCESS.2020.3000179). URL: <https://doi.org/10.1109/ACCESS.2020.3000179>.
- [4] Jesús Díaz-Verdejo et al. “On the Detection Capabilities of Signature-Based Intrusion Detection Systems in the Context of Web Attacks”. In: *Applied Sciences* 12.2 (2022), p. 852.
- [5] Mohamed Amine Ferrag et al. “Edge-IIoTset: A New Comprehensive Realistic Cyber Security Dataset of IoT and IIoT Applications for Centralized and Federated Learning”. In: *IEEE Access* 10 (2022), pp. 40281–40306. DOI: [10.1109/ACCESS.2022.3165809](https://doi.org/10.1109/ACCESS.2022.3165809). URL: <https://doi.org/10.1109/ACCESS.2022.3165809>.
- [6] Abdullah Bin Jasni, Akiko Manada, and Kohei Watabe. “DiffuPac: Contextual Mimicry in Adversarial Packets Generation via Diffusion Model”. In: *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*. Ed. by Amir Globersons et al. 2024. URL: http://papers.nips.cc/paper%5C_files/paper/2024/hash/f1a92c4df8cd7dc1cab2613fb999d5e7-Abstract-Conference.html.
- [7] Yaokai Feng and Kouichi Sakurai. “Network Intrusion Detection: Evolution from Conventional Approaches to LLM Collaboration and Emerging Risks”. In: *CoRR* abs/2510.23313 (2025). DOI: [10.48550/ARXIV.2510.23313](https://doi.org/10.48550/ARXIV.2510.23313). arXiv: [2510.23313](https://arxiv.org/abs/2510.23313). URL: <https://doi.org/10.48550/arXiv.2510.23313>.
- [8] Peishuai Sun et al. “AdvTG: An Adversarial Traffic Generation Framework to Deceive DL-Based Malicious Traffic Detection Models”. In: *Proceedings of the ACM on Web Conference 2025, WWW 2025, Sydney, NSW, Australia, 28 April 2025- 2 May 2025*. Ed. by Guodong Long et al. ACM, 2025, pp. 3147–3159. DOI: [10.1145/3696410.3714876](https://doi.org/10.1145/3696410.3714876). URL: <https://doi.org/10.1145/3696410.3714876>.